

Exploring the Database Structure

Experienced SQL users can often use database queries to infer the structure of a database. But each database system has different ways of managing this information. The SQL Murder Mystery is built using SQLite. Use this SQL command to find the tables in the Murder Mystery database.

Run this query to find the names of the tables in this database.

This command is specific to SQLite. For other databases, you'll have to learn their specific syntax.

```
1 SELECT name
2 FROM sqlite_master
3 where type = 'table'
```

RUN ↴

RESET

name
crime_scene_report
drivers_license
facebook_event_checkin
interview
get_fit_now_member
get_fit_now_check_in
solution
income
person

Besides knowing the table names, you need to know how each table is structured. The way this works is also dependent upon which database technology you use. Here's how you do it with SQLite.

Run this query to find the structure of the `crime\_scene\_report` table

Change the value of 'name' to see the structure of the other tables you learned about with the previous query.

```
1 SELECT sql
2 FROM sqlite_master
3 where name = 'crime_scene_report'
```

RUN ↴

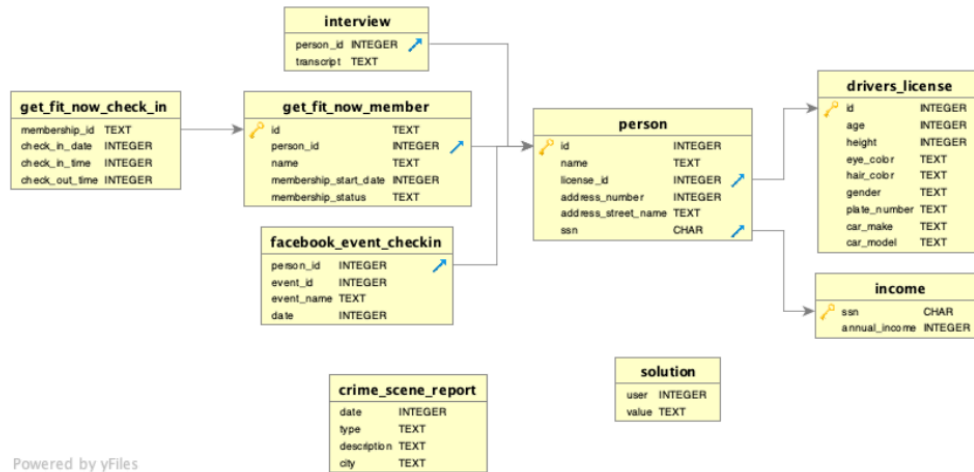
RESET

sql
CREATE TABLE crime_scene_report (date integer, type text, description text, city text)

The rest is up to you!

If you're really comfortable with SQL, you can probably get it from here.

But [click here](#) to show the schema diagram.



And you can always go to the [walkthrough](#).

Use your knowledge of the database schema and SQL commands to find out who committed the murder.

When you think you know the answer, go to the next section.

```
1 SELECT description
2 FROM crime_scene_report
3 WHERE CITY = 'SQL City'
4     AND type = 'murder'
5     AND date = 20180115;
```

RUN ↴

RESET

description

Security footage shows that there were 2 witnesses. The first witness lives at the last house on "Northwestern Dr". The second witness, named Annabel, lives somewhere on "Franklin Ave".

The next logical step is to find each witness.

Use your knowledge of the database schema and SQL commands to find out who committed the murder.

When you think you know the answer, go to the next section.

```
1 SELECT
2     id,
3     name,
4     address_number
5 FROM person
6 WHERE address_street_name = 'Northwestern Dr'
7 LIMIT 1
8
9
10
11
12
```

RUN ↴

RESET

id	name	address_number
10010	Muoi Cary	741

Witness 1: Muoi Cary at 741 Northwestern Dr, ID number 10010

when you think you know the answer, go to the next section.

```
1 SELECT
2     id,
3     name,
4     address_number
5 FROM person
6 WHERE address_street_name = 'Franklin Ave'
7       AND name LIKE '%Annabel%'
8 LIMIT 1
9
10
11
12
13
```

RUN ↴

RESET

id	name	address_number
16371	Annabel Miller	103

Witness 2: Annabel Miller at 103 Franklin Ave, ID number 16371

Now that the witnesses have been identified, I must use their ID numbers to locate their interview transcripts.

when you think you know the answer, go to the next section.

```
1 SELECT
2     person_id,
3     transcript
4 FROM interview
5 WHERE person_id = 10010
6
7
8
9
```

This gave me no data returned. I looked and realized it's because I found the wrong witness. I missed "last house," thinking it wasn't important. Here's the correct syntax for that witness:

Use your knowledge of the database schema and SQL commands to find out who committed the murder.

When you think you know the answer, go to the next section.

```
1 SELECT
2     id,
3     name,
4     address_number
5 FROM person
6 WHERE address_street_name = 'Northwestern Dr'
7 ORDER BY address_number DESC
8 LIMIT 1;
9
10
11
12
```

RUN ↴

RESET

id	name	address_number
14887	Morty Schapiro	4919

Now that I have both correct witnesses, here's the syntax:

Use your knowledge of the database schema and SQL commands to find out who committed the murder.

When you think you know the answer, go to the next section.

```
1 SELECT
2   person_id,
3   transcript
4 FROM interview
5 WHERE person_id IN (14887, 16371);
6
7
8
```

RUN ↴

RESET

person_id	transcript
14887	I heard a gunshot and then saw a man run out. He had a "Get Fit Now Gym" bag. The membership number on the bag started with "48Z". Only gold members have those bags. The man got into a car with a plate that included "H42W".
16371	I saw the murder happen, and I recognized the killer from my gym when I was working out last week on January the 9th.

This doesn't seem to be the correct syntax. I'll have to adjust:

Use your knowledge of the database schema and SQL commands to find out who committed the murder.

When you think you know the answer, go to the next section.

```
1 SELECT
2   gtnm.person_id,
3   gtnm.name,
4   dl.id,
5   dl.plate_number
6 FROM get_fit_now_member gtnm
7 INNER JOIN drivers_license dl ON gtnm.id = dl.id
8
9
10
```

RUN ↴

RESET

No data returned

I had to adjust the syntax and add another inner join. I looked up the plate number along with their name. Got a suspect:

When you think you know the answer, go to the next section.

```
1 SELECT
2     gtnm.person_id,
3     gtnm.name,
4     dl.plate_number
5 FROM get_fit_now_member gtnm
6 INNER JOIN person p ON gtnm.person_id = p.id
7 INNER JOIN drivers_license dl ON p.license_id = dl.id
8 WHERE dl.plate_number LIKE '%H42W%';
9
10
11
12
```

RUN ↴ RESET

person_id	name	plate_number
67318	Jeremy Bowers	0H42W2

Next, I need to look up the Get Fit Now Check In to see if Jeremy Bowers was there on January 9, the day of the murder.

```
1 SELECT
2     ci.membership_id,
3     m.name,
4     ci.check_in_date,
5     ci.check_in_time,
6     ci.check_out_time
7 FROM get_fit_now_check_in ci
8 INNER JOIN get_fit_now_member m ON ci.membership_id = m.id
9 WHERE m.name = 'Jeremy Bowers'
10 AND ci.check_in_date = '01092018'
11
12
13
14
```

RUN ↴ RESET

No data returned

This yield no data, probably because the date is not in the right format. I'm going to look up the check out date data to see the format.

Use your knowledge of the database schema and SQL commands to find out who committed the murder.
When you think you know the answer, go to the next section.

```
1 SELECT *
2 FROM get_fit_now_check_in
3
4
5
6
```

RUN ↴

RESET

membership_id	check_in_date	check_in_time	check_out_time
NL318	20180212	329	365
NL318	20170811	469	920
NL318	20180429	506	554
NL318	20180128	124	759
NL318	20171027	418	1019
NL318	20171018	608	1074
NL318	20171127	850	954

With the right format, I can now run the original code with the date fixed.

Use your knowledge of the database schema and SQL commands to find out who committed the murder.

When you think you know the answer, go to the next section.

```
1 SELECT
2     ci.membership_id,
3     m.name,
4     ci.check_in_date,
5     ci.check_in_time,
6     ci.check_out_time
7 FROM get_fit_now_check_in ci
8 INNER JOIN get_fit_now_member m ON ci.membership_id = m.id
9 WHERE m.name = 'Jeremy Bowers'
10    And ci.check_in_date = '20180109';
11
12
13
14
```

RUN ↴

RESET

membership_id	name	check_in_date	check_in_time	check_out_time
48Z55	Jeremy Bowers	20180109	1530	1700

This confirms that Jeremy Bowers is who one of the witnesses saw leaving the gym. Now I'm going to look up the interview that the police did with him, if it's there.

Use your knowledge of the database schema and SQL commands to find out who committed the murder.

When you think you know the answer, go to the next section.

```
1 SELECT
2     p.name,
3     i.transcript
4 FROM interview i
5 INNER JOIN person p ON i.person_id = p.id
6 WHERE p.name = 'Jeremy Bowers'
7
8
9
```

RUN ↴

RESET

name	transcript
Jeremy Bowers	I was hired by a woman with a lot of money. I don't know her name but I know she's around 5'5" (65") or 5'7" (67"). She has red hair and she drives a Tesla Model S. I know that she attended the SQL Symphony Concert 3 times in December 2017.

Based on this interview, I need to find the name of the woman Jeremy described in the interview.


```

1 SELECT
2     p.name,
3     dl.height,
4     dl.hair_color,
5     dl.gender,
6     dl.car_make,
7     dl.car_model
8 FROM drivers_license dl
9 INNER JOIN person p ON dl.id = p.license_id
10 WHERE dl.car_make = 'Tesla'
11     AND dl.car_model = 'Model S'
12     AND dl.height = '65"' OR '67"'
13
14
15
16
17
18
19

```

RUN ↴

RESET

name	height	hair_color	gender	car_make	car_model
Christopher Peteuil	59	green	male	Chrysler	Town & Country
Kourtney Calderwood	74	white	female	BMW	M Roadster
Muoi Cary	79	green	female	Mercedes-Benz	CLS-Class
Era Moselle	75	red	female	Ford	Ranger
Trena Hornby	60	black	female	Toyota	Tacoma Xtra
Antione Godbolt	72	blue	male	Honda	Civic
Kyra Buen	69	grey	female	Infiniti	G
Francesco Agundez	54	blue	male	Pontiac	Grand Am
Leslie Thate	74	blonde	female	GMC	Yukon

I tried to find the specific person, but I didn't properly restrict the conditions. After doing some research, I think I need to put parentheses around my last statement after AND and get rid of the quotes.

When you think you know the answer, go to the next section.

```
1 SELECT
2     p.name,
3     dl.height,
4     dl.hair_color,
5     dl.gender,
6     dl.car_make,
7     dl.car_model
8 FROM drivers_license dl
9 INNER JOIN person p ON dl.id = p.license_id
10 WHERE dl.car_make = 'Tesla'
11        AND dl.car_model = 'Model S'
12        AND (dl.height = 65 OR 67)|
13
14
15
16
17
18
19
```

RUN ↴

RESET

name	height	hair_color	gender	car_make	car_model
Dion Montague	65	white	male	Tesla	Model S
Cletus Zoeller	69	red	male	Tesla	Model S
Alyssa Faggett	80	grey	female	Tesla	Model S
Red Korb	65	red	female	Tesla	Model S
Regina George	66	red	female	Tesla	Model S
Jon Haldi	63	white	male	Tesla	Model S
Miranda Priestly	66	red	female	Tesla	Model S

I narrowed it down more, but I realized I still need to add hair color and gender to narrow it down further.

Use your knowledge of the database schema and SQL commands to find out who committed the murder.

When you think you know the answer, go to the next section.

```
1 SELECT
2     p.name,
3     dl.height,
4     dl.hair_color,
5     dl.gender,
6     dl.car_make,
7     dl.car_model
8 FROM drivers_license dl
9 INNER JOIN person p ON dl.id = p.license_id
10 WHERE dl.car_make = 'Tesla'
11        AND dl.car_model = 'Model S'
12        AND dl.height IN (65,67)
13        AND dl.hair_color = 'red'
14        AND dl.gender = 'female'
15
16
17
18
19
20
21
```

RUN ↴

RESET

name	height	hair_color	gender	car_make	car_model
Red Korb	65	red	female	Tesla	Model S

Now I need to find her interview transcript. However, I realized I don't have her ID. I need to add it to the query.

```

1 SELECT
2     p.id AS person_id,
3     p.name,
4     dl.height,
5     dl.hair_color,
6     dl.gender,
7     dl.car_make,
8     dl.car_model
9 FROM drivers_license dl
10 INNER JOIN person p ON dl.id = p.license_id
11 WHERE dl.car_make = 'Tesla'
12     AND dl.car_model = 'Model S'
13     AND dl.height IN (65,67)
14     AND dl.hair_color = 'red'
15     AND dl.gender = 'female'
16
17
18
19
20
21
22

```

RUN ↴

RESET

person_id	name	height	hair_color	gender	car_make	car_model
78881	Red Korb	65	red	female	Tesla	Model S

Now I'm ready to find the interview.

```

1 SELECT
2     p.name,
3     i.person_id,
4     i.transcript
5 FROM interview i
6 INNER JOIN person p ON p.id = i.person_id
7 WHERE name = 'Red Korb'
8
9
10
11
12
13
14

```

RUN ↴

RESET

No data returned

I think I made the wrong connection. I'm going to adjust and try again. It turns out that Red Korb isn't the person. I made an error back when finding the driver. I'll need to adjust that.

When you think you know the answer, go to the next section.

```
1 SELECT
2     p.id AS person_id,
3     p.name,
4     dl.height,
5     dl.hair_color,
6     dl.gender,
7     dl.car_make,
8     dl.car_model
9 FROM drivers_license dl
10 INNER JOIN person p ON dl.id = p.license_id
11 WHERE dl.car_make = 'Tesla'
12        AND dl.car_model = 'Model S'
13        AND dl.height BETWEEN 65 AND 67
14        AND dl.hair_color = 'red'
15        AND dl.gender = 'female';
16
17
18
19
20
21
22
23
```

RUN ↴

RESET

person_id	name	height	hair_color	gender	car_make	car_model
78881	Red Korb	65	red	female	Tesla	Model S
90700	Regina George	66	red	female	Tesla	Model S
99716	Miranda Priestly	66	red	female	Tesla	Model S

After analyzing Jeremy's earlier transcript, I realized the height was between 65 and 67. So I've narrowed it down to three women. I also remember he mentioned a concert. I think that is the Facebook event. So I'll use that to narrow it down.

```

1 SELECT
2     p.name,
3     COUNT(f.event_id) AS times_attended
4 FROM person p
5 INNER JOIN facebook_event_checkin f ON p.id = f.person_id
6 WHERE p.name IN ('Red Korb', 'Regina George', 'Miranda Priestly')
7     AND f.event_name = 'SQL Symphony Concert'
8     AND f.date LIKE '201712%'
9 GROUP BY p.name;
10
11
12
13
14
15
16
17

```

RUN ↴

RESET

name	times_attended
Miranda Priestly	3

This has narrowed it down to Miranda Priestly. It matches what Jeremy said in his interview. Now I am going to see if her name is the solution to the murder mystery.

Did you find the killer?

```

1 INSERT INTO solution VALUES (1, 'Miranda Priestly');
2
3     SELECT value FROM solution;

```

RUN ↴

RESET

value

Congrats, you found the brains behind the murder! Everyone in SQL City hails you as the greatest SQL detective of all time. Time to break out the champagne!

I successfully found the person behind the murder, Miranda Priestly!